Teacher Guide: Week 1

Logic Is Not Magic: What Logic Can and Cannot Do

12th Grade Long Logic Unit Based on Richard Whately's *Elements of Logic*

Week 1 Overview

This first week introduces Richard Whately's basic account of logic for a twelfth-grade long unit. Students should leave the week understanding that logic is not a machine that automatically discovers truth. Logic studies the structure of reasoning and helps expose bad inferences. The reading teaches the pattern of conclusion, stated reason, hidden assumption, and test; the argument lab then asks students to transfer that pattern to fresh examples and to the recurring Shakespeare anchors, *Macbeth* and *Julius Caesar*. This establishes the foundation for later weeks on terms, propositions, syllogisms, mood and figure, fallacies, ambiguity, induction, rhetoric, and verbal versus real disputes.

Essential Question

What can logic actually do for a thinker, and what should we not expect it to do?

Student-Facing Materials

- *Logic Is Not Magic: What Logic Can and Cannot Do* — student reading handout.
- *Week 1 Argument Lab: What Follows From What?* — transfer-practice handout with fresh argument tests, a repair task, an original argument task, and an exit claim.
- *Macbeth Lit-Anchor: Prophecy Is Not Proof* — Shakespeare anchor passage for applying Whately's four questions to Macbeth's movement from prophecy to inference, ambition, and moral responsibility.

Learning Goals

Students will be able to:

1. Define logic as the study of reasoning and argument structure.
2. Distinguish what logic can do from what it cannot do.
3. Identify conclusion, stated reason, and hidden assumption in a simple argument.
4. Explain why popularity, public honor, tradition, difficult language, confidence, or emotion do not by themselves prove a claim.
5. Repair a weak argument by narrowing the claim or strengthening the support.

6. Create and test an original argument.
7. Apply the four-step test to an initial *Macbeth* or *Julius Caesar* claim for the course portfolio.

Key Vocabulary

logic, reasoning, argument, premise, conclusion, deduction, fallacy, science, art, panacea, hidden assumption

Weekly Lesson Sequence

Day 1: Opening Question and First Read

- Begin with the question: *If someone is good at arguing, does that mean he is good at finding truth?*
- Have students answer briefly in writing.
- Introduce Whately as a nineteenth-century thinker trying to rescue logic from both overpraise and contempt.
- Read the introduction and first half of the student reading aloud or in pairs.
- Stop after the analogy paragraph. Ask: What does Whately think logic is being unfairly expected to do?

Day 2: Finish Reading and Annotate Whately's Claim

- Finish the reading.
- Students mark three places: one sentence showing what logic can do, one showing what it cannot do, and one example that separates conclusion, stated reason, hidden assumption, and test.
- Complete Reading Focus questions 1–6.

Day 3: Argument Skeleton Practice

- Introduce the pattern: conclusion, stated reason, hidden assumption, test.
- Work through the modeled popularity/fairness example from the Argument Lab handout. Name it explicitly as the one repeated example from the reading.
- Students complete the fresh Part II arguments individually, then compare answers with a partner.
- Tell students that the goal is transfer: the reading already showed the method, but the lab asks whether they can use it on new arguments.
- Emphasize that Whately's basic question is: *What follows from what?*

Day 4: Macbeth Lit-Anchor: Prophecy Is Not Proof

- Read the guided Act 1, Scene 3 excerpt from the lit-anchor: the witches' greeting, Ross's confirmation of Cawdor, Banquo's warning, and Macbeth's first aside.
- Build a four-question skeleton on the board: What conclusion does Macbeth begin to draw? What reason does he have? What hidden assumption links prophecy to action? Does the conclusion follow?
- Keep the key distinction visible: the witches give a prediction, not a command. Macbeth adds interpretation, desire, and possible action.
- Have students read the Act 1, Scene 7 independent passage and begin the *Macbeth Argument Skeleton* portfolio artifact.

Day 5: Repair, Portfolio, and Exit Claim

- Students finish either the repaired argument/original test-case work from the lab or the *Macbeth Argument Skeleton* from the lit-anchor.
- Require one reference to Whately's view and one example from school, literature, public speech, everyday life, or *Macbeth*.
- Use Macbeth to synthesize the week: logic can expose a bad inference, but it cannot make Macbeth virtuous.
- Collect the student reading, argument lab, and lit-anchor portfolio artifact as the week's work product.

Lit-Anchor Teaching Notes

The Week 1 lit-anchor is not a plot worksheet. Its purpose is to let students use Whately's modest account of logic on a familiar literary problem: Macbeth hears a prophecy, then begins making inferences from it. Keep students from saying simply, "the witches made him do it." The more precise Whately-style question is whether Macbeth's conclusions follow from the reasons available to him.

- **Guided passage:** *Macbeth* Act 1, Scene 3, from the witches' greeting through Macbeth's "If chance will have me king" aside.
- **Independent passage:** Act 1, Scene 7, Macbeth's soliloquy before Duncan's murder.
- **Central logical pressure:** prediction is not permission; possibility is not command; desire is not proof.
- **Portfolio artifact:** Macbeth argument skeleton: claim/conclusion, stated reason, hidden assumption, and test.

A strong class discussion should separate three claims: (1) Macbeth may become king; (2) Macbeth should act to become king; (3) murder may be a legitimate path to kingship. The first has some support once Cawdor is confirmed. The second and third outrun the evidence. Banquo's warning is important because it offers a rival interpretation: truthful fragments can still be bait.

Suggested Board Notes

- Logic does not discover every truth.
- Logic studies whether a conclusion follows from premises.
- Bad facts can still produce bad conclusions even when the reasoning looks neat.
- A reading example may teach the method; fresh examples test whether students can transfer it.
- Popularity, public honor, tradition, difficult language, confidence, and repetition are not the same as proof.
- The first logical question: What follows from what?

Argument Lab Structure

- **Part I** is a model from the reading. Do not treat it as the main challenge; use it to rehearse the four-part method.
- **Part II** contains four fresh arguments. Arguments 1–3 are deliberately weak transfer cases. Argument 4 is stronger, but only if the two students' situations are truly alike in all relevant ways.
- **Part III** asks students to repair one weak argument. Accept either a more careful conclusion or added support that actually bears on the conclusion.
- **Part IV** asks students to create an original argument and test their own reasoning.
- The exit claim asks students to explain when logic helps a reader most, not merely to praise logic in general.

Reading Focus Answer Guide

1. Main claim: Logic is the science and art of reasoning; it studies argument structure and helps guard against bad deductions.
2. Can do: examine whether conclusions follow; expose skipped steps, changed terms, hidden assumptions, or bad inferences.
3. Cannot do: supply facts by itself; make foolish people wise; discover every truth; replace observation, experience, memory, imagination, or moral judgment.

4. Four steps: find the conclusion, find the stated reason, identify the hidden assumption, and test whether the conclusion really follows.
5. Answers will vary. Strong responses should use one reading example and correctly name its conclusion, reason, assumption, and test. For example, the Macbeth argument concludes that Macbeth is not responsible; its stated reason is the witches' prediction; its hidden assumption is that prediction removes responsibility; the conclusion does not necessarily follow. In the lit-anchor, students may also test Macbeth's more immediate inference that one fulfilled prediction makes kingship certain or that a predicted future authorizes action.
6. People blamed logic because some defenders promised too much for it, so critics rejected it when it failed to perform impossible tasks.

Argument Lab Answer Guide

The revised lab keeps one modeled argument from the reading, then moves students into fresh test cases. Strong student answers should not merely label parts; they should explain whether the reason actually earns the conclusion.

Model From the Reading: Popularity and Fairness

- Conclusion: The rule must be fair.
- Stated reason: Most students like it.
- Hidden assumption: Whatever most students like is fair.
- Test: The conclusion does not necessarily follow. Popularity may sometimes accompany fairness, but popularity alone does not prove fairness.

Argument 1: Statue and Noble Life

- Conclusion: The man must have lived nobly.
- Stated reason: The city placed a bronze statue of him in the town square.
- Hidden assumption: Public honor always goes to people who lived nobly, or a statue proves virtue.
- Test: The conclusion does not necessarily follow. A statue proves that a community chose to honor someone; it does not by itself prove that the person deserved the honor.

Argument 2: Difficult Words and Better Reasoning

- Conclusion: The first essay must have better reasoning.
- Stated reason: The first essay uses more difficult words than the second essay.
- Hidden assumption: More difficult language means better reasoning.

- Test: The conclusion does not follow. Difficult words may hide weak reasoning, and plain words may express strong reasoning.

Argument 3: Always Done This Way

- Conclusion: The old method must be the best method.
- Stated reason: The class has always done the assignment this way.
- Hidden assumption: Whatever has been done for a long time is best.
- Test: The conclusion does not necessarily follow. Tradition may preserve wisdom, but length of use alone does not prove superiority.

Argument 4: Same Action, Different Punishment

- Conclusion: The rule is not just.
- Stated reason: A just rule should treat the same action the same way, but this rule punishes one student and excuses another for the same action.
- Hidden assumption: Relevant circumstances are truly the same; no morally important difference justifies the different treatment.
- Test: This is a stronger argument if the cases really are alike in all relevant ways. If important circumstances differ, the argument needs more information.

Repair and Original Test Case

- Repaired arguments should either narrow an overclaim or add a reason that directly supports the conclusion.
- Original student arguments should include a clear conclusion and a stated reason. Stronger responses will identify a real hidden assumption rather than repeating the stated reason.

Assessment

Use the repaired argument, original test case, Shakespeare application, and portfolio exit paragraph as formative assessment. Strong work should:

- identify conclusions and reasons accurately;
- state hidden assumptions in complete, testable form;
- explain whether the reason actually earns the conclusion;
- improve a weak argument by narrowing the claim or strengthening the support;
- avoid treating impressive language, tradition, honor, or confidence as automatic proof.

Next Week Preview

Week 2 should move from reasoning in general to **terms**: the words that carry arguments. The next guiding question should be: *Why do arguments become unstable when key words are unclear?* Keep returning to *Macbeth* and *Julius Caesar* so students can practice new logic tools on familiar literary ground.